



EE 580: Network Systems: Modelling and Analysis

Course Code	EE 580
Course Name	Network Systems: Modelling and Analysis
L-T-P-C	3-0-0-3
Prerequisites	MA 512 Linear Algebra or EE 522 Matrix theory for engineers and EE509 Linear Dynamical Systems or instructor's consent
Intended for	3rd/ 4th year B.Tech. (all branches), MSc., M.Tech, Ph.D.
Distribution	Discipline elective for B.Tech. (EE, CSE), free elective for others
Approval	recommended by the 35th BoA (Aug 2020)

Course Contents

- **Introduction:** Examples of networks systems, engineered (sensor & robotic networks) and natural (social & compartmental networks). Need for tools to understand and analyse these systems. [2 Lectures]
- **Review of Matrix theory & Graph theory:** Jordan normal form, stochastic matrices and spectral radius, Perron-Frobenius theory. Graphs and digraphs, weighted digraphs. [4 Lectures]
- **Algebraic graph theory:** Adjacency matrix, graph theoretical characterization of primitive & irreducible matrices. Elements of spectral graph theory. [4 Lectures]
- **Averaging systems:** Discrete-time and continuous-time averaging systems. Consensus in averaging systems. Convergence properties. Laplacian flow. Design of weighted digraphs. Scalability, optimization. Time-varying and randomized averaging algorithms. Examples: sensor networks, opinion dynamics over social networks. [10 Lectures]
- **Compartmental systems:** Positive systems and Metzler matrices. Compartmental matrices. Dynamic properties. Spectral properties. Algebraic and graphical properties. Examples: epidemiology, drug kinetics. [8 Lectures]
- **Stability theory for networks:** Dynamical systems and stability notions, Lyapunov stability criteria, Krasovskii-LaSalle invariance principle. Linear, nonlinear and linearized systems. Negative gradient systems. [7 Lectures]
- **Introduction to simulation:** Motivation - role of simulation in understanding and engineering network systems. Discussion on available tools MATLAB/Simulink, NetSim, NS2. Design of simulation experiments for sensor networks, computations on models for compartmental systems. [3 Lectures]
- **Advanced topics:** Multi-agent systems, Wireless sensor and actuators networks, robot swarms. Problems and trends. [3 Lectures]

Text Books

1. Francesco Bullo, **Lectures on Network Systems**, CreateSpace, 2018.
2. Albert-Laszlo Barabasi, **Network Science**, Cambridge University Press, 2016.

Additional References

1. Mark Newman, Albert-Laszlo Barabasi and Duncan J. Watts, **The structure and dynamics of networks**, Princeton university press, 2006.
2. Mark Newman, **Networks An Introduction**, Oxford University Press, 2018.
3. Guanrong Chen, Xiaofan Wang, and Xiang Li, **Fundamentals of complex networks: models, structures and dynamics**, John Wiley & sons, 2014.
4. Selected publications to be chosen by the instructor.